SATELLITE PHASED-ARRAY ANTENNA PERFORMANCE SIMULATOR

Automated Engineering Verification Report

Report generated: 2026-11-04 19:56:35

Test environment: Python-based theoretical simulation

Important: This report presents results from an educational mathematical antenna simulation. No physical antenna, RF instrument, or satellite hardware was tested. The model is not validated against MDA AURORA or IRIS2 hardware.

1. Project Overview

This project implements a theoretical planar phased-array antenna model. The simulator calculates element positions, steering phases, radiation patterns, and selected antenna performance metrics. It also evaluates simulated component failures and phase errors through fault-injection scenarios. Frequency and beam-steering sweeps are used to examine how selected performance metrics change as operating conditions vary.

2. Antenna Configuration

Parameter	Configured Value
Operating frequency	20.00 GHz
Wavelength	14.990 mm
Array dimensions	16 x 16
Total elements	256
Element spacing X	7.495 mm
Element spacing Y	7.495 mm
Commanded theta	30.00 degrees
Commanded phi	0.00 degrees
Element pattern exponent	1.0

3. Baseline Antenna Performance

Performance Metric	Simulated Result
Commanded beam direction	30.000 degrees
Peak beam direction	29.896 degrees
Pointing error	0.104 degrees
Half-power beamwidth	7.324 degrees
Peak sidelobe level	-12.76 dB
Left -3 dB crossing	26.315 degrees
Right -3 dB crossing	33.639 degrees

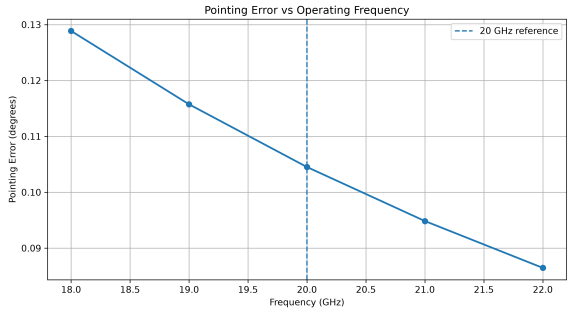
Measurements are calculated from the simulated radiation pattern. Pointing error and sidelobe results are specific to the selected angular sampling and the phi = 0 degree cut.

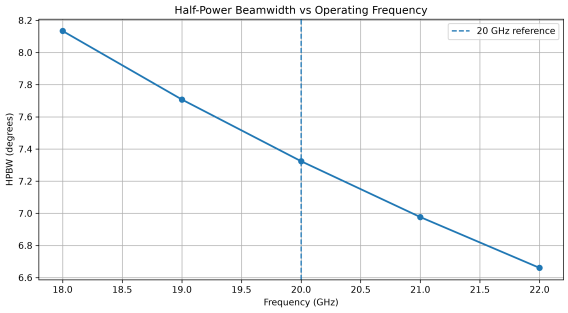
4. Frequency Sweep Analysis

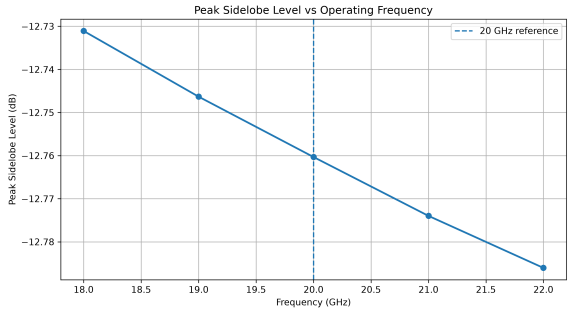
The frequency sweep evaluates the simulated antenna response at five operating frequencies between 18 and 22 GHz. The physical element positions and spacing remain fixed at the reference design values. Steering phases are recalculated at each frequency to maintain the commanded beam direction.

Frequency (GHz)	Peak angle (deg)	Pointing error (deg)	HPBW (deg)	Peak SLL (dB)
18.00	29.871	0.129	8.134	-12.73
19.00	29.884	0.116	7.708	-12.75
20.00	29.895	0.105	7.324	-12.76
21.00	29.905	0.095	6.977	-12.77
22.00	29.914	0.086	6.661	-12.79

Frequency sweep results and performance plots:







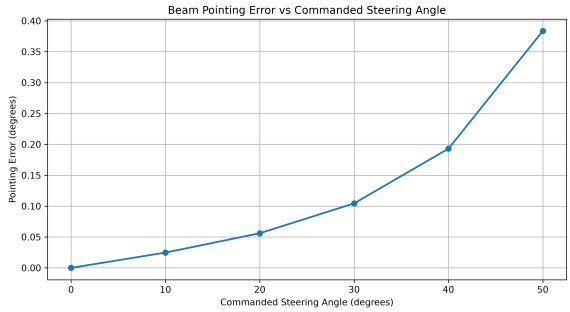
This is a phase-adjusted frequency sweep, not a fixed-phase beam-squint experiment. It does not establish measured frequency-dependent antenna performance.

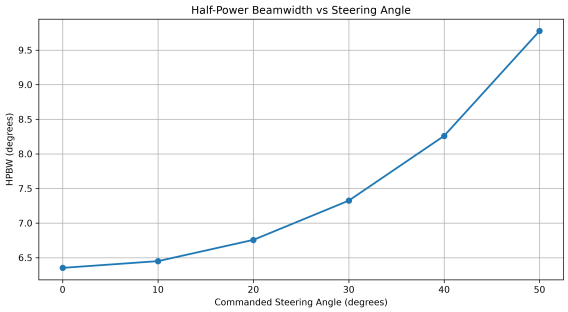
5. Beam-Steering Sweep Analysis

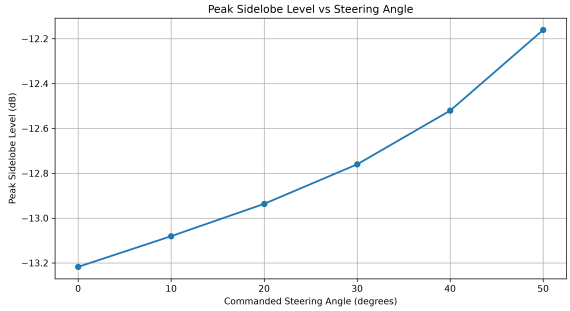
The beam-steering sweep evaluates the antenna at commanded steering angles from 0 to 50 degrees, in 10-degree increments. Operating frequency and physical element spacing remain fixed. Parabolic interpolation is applied around the sampled main-beam maximum to estimate its peak direction between angular samples.

Command (deg)	Peak angle (deg)	Pointing error (deg)	HPBW (deg)	Peak SLL (dB)
0.0	-0.0000	0.0000	6.352	-13.22
10.0	9.9752	0.0248	6.450	-13.08
20.0	19.9439	0.0561	6.757	-12.94
30.0	29.8955	0.1045	7.324	-12.76
40.0	39.8070	0.1930	8.260	-12.52
50.0	49.6163	0.3837	9.776	-12.16

Beam-steering sweep results and performance plots:



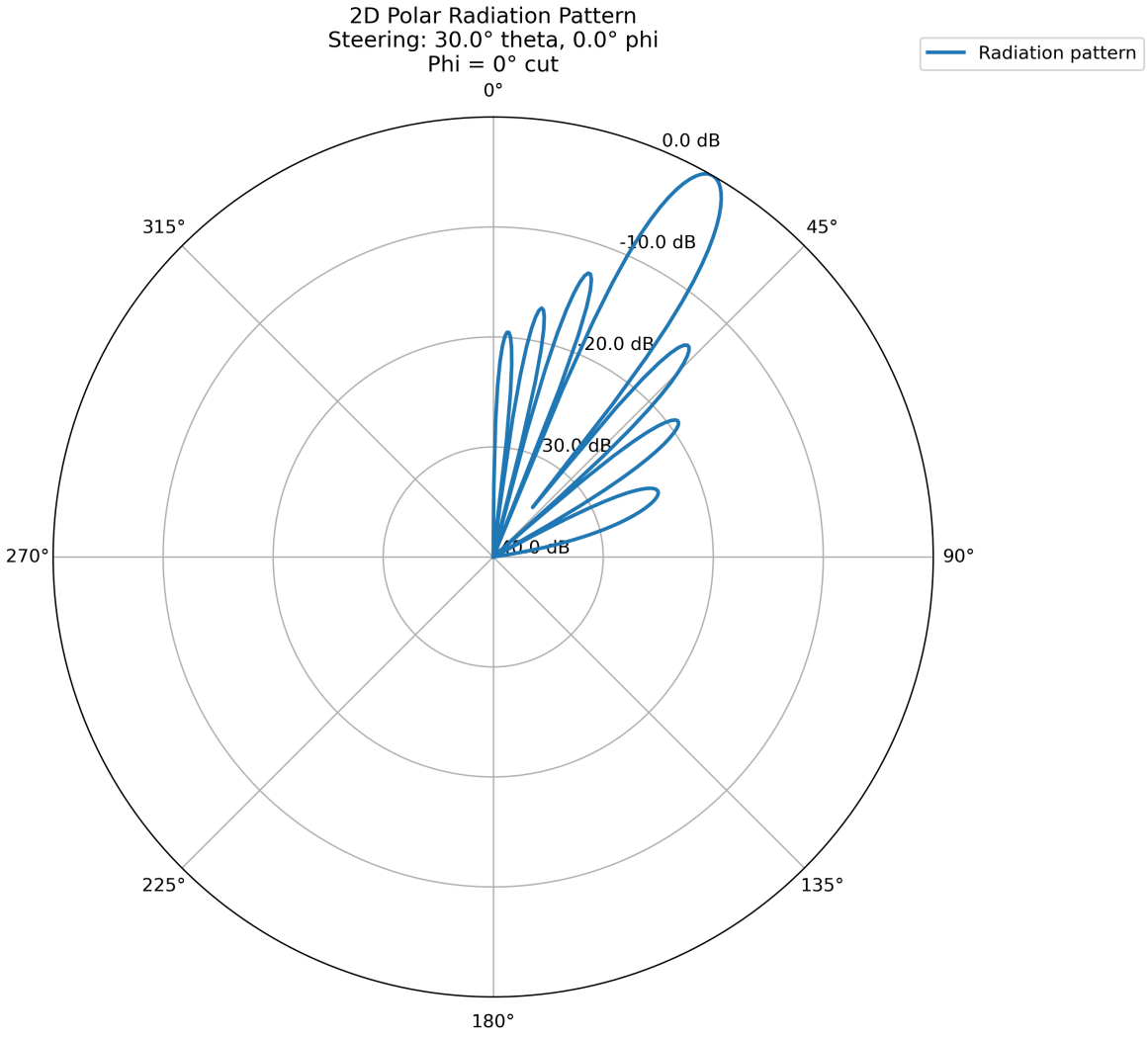


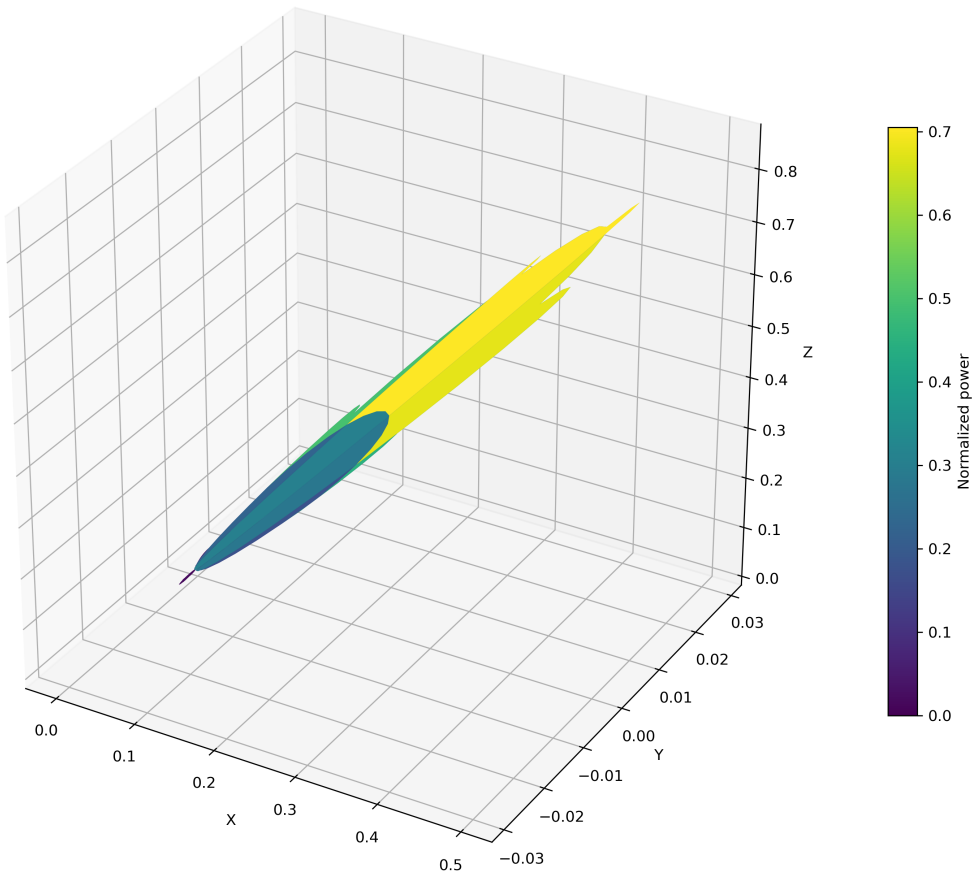


The interpolated peak is a numerical estimate based on the simulated angular pattern. It is not a physical measurement or a guarantee of antenna pointing accuracy.

6. Radiation Pattern Visualizations

The following figures illustrate the simulated antenna radiation characteristics for the baseline configuration.

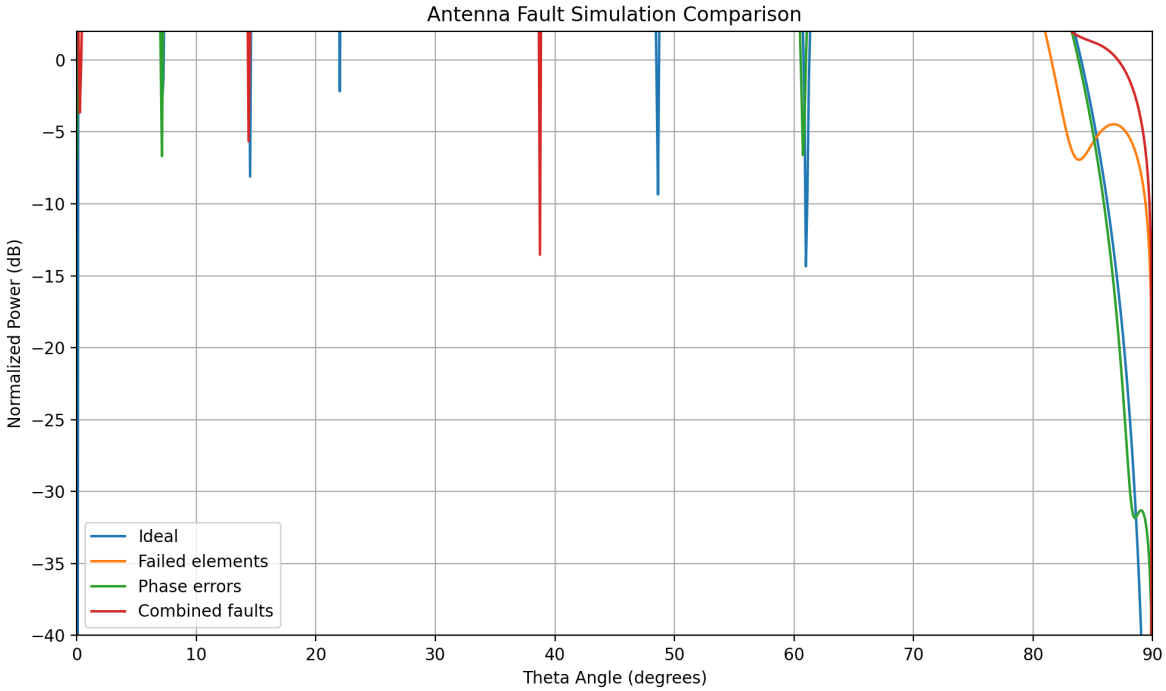



3D Satellite Phased-Array Radiation Pattern
Steering angle: 30.0°
X
Y
Z
Normalized power
0.0
0.1
0.2
0.3
0.4
0.5
−0.03
−0.02
−0.01
0.00
0.01
0.02
0.03
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7

7. Fault Injection Analysis

The failed-element scenario disabled 256 of 256 antenna elements (100.00%). Random phase errors were also introduced to evaluate their effects on simulated peak power. The phase-error scenario uses a reproducible random seed.

Fault Scenario	Relative Peak Power
Ideal antenna	0.00 dB
Failed elements	-0.93 dB
Phase errors	-0.12 dB
Combined faults	-1.04 dB



Relative peak power is compared with the ideal simulated pattern. These scenarios represent software-injected faults, not faults measured on physical antenna hardware.

8. Automated Requirements Verification

Requirement	Measured Result	Criterion	Status
			PASS
			PASS
			PASS
			PASS
			PASS
			PASS

9. Verification Conclusion

The automated verification process completed with 6 of 6 project-defined acceptance checks passing.

Limitations: The simulator uses an idealized mathematical array model and an assumed element radiation pattern. It does not include full-wave electromagnetic analysis, hardware tolerances, measured antenna data, environmental testing, or validation against an actual satellite antenna. The frequency sweep recalculates steering phases at each frequency. Sidelobe analysis is based on the phi = 0 degree pattern cut. All acceptance criteria are educational project requirements, not manufacturer specifications.